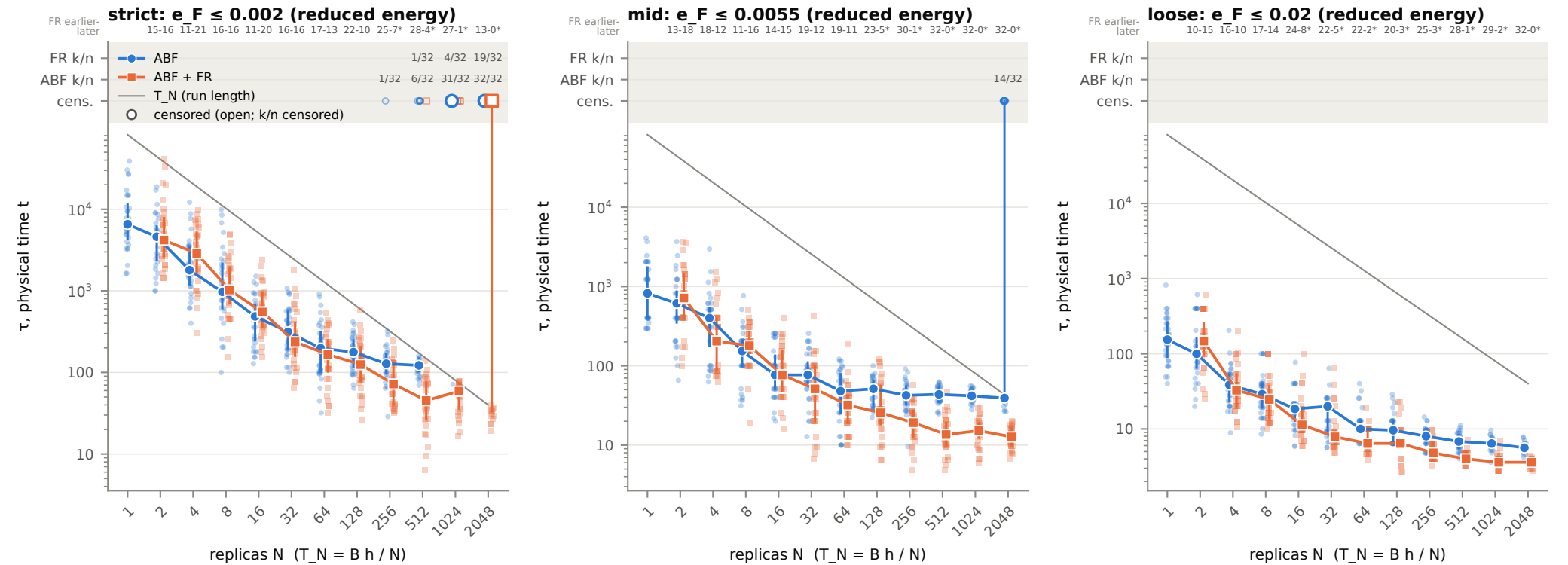


# entropic gateway: persistent time to free-energy accuracy, physical time

Equal total budget per arm per seed:  $B = N \times n_{\text{steps}} = 3.2768\text{e}+09$  walker-steps at every  $N$  ( $h = 2.5\text{e}-05$ ,  $T_N = B \cdot h / N$ ); 32 seeds per ( $N$ , method).  $\tau(\epsilon)$  = first save after which  $e_F \leq \epsilon$  at every later save; never met: censored ( $> T_N$ ,  $> 1$ ).



entropic gateway · S4a\_tau\_free\_energy\_time · status: all planned ( $N$ , seed, method) complete · FR is not defined at  $N = 1$  (ABF only, by design).  
Markers: seed median (large open marker in the 'cens.' band = median censored); bars: IQR; faint: seeds; censored seeds are open markers in the band, never success at  $T_N$ .  
Row above each panel: the PAIRED comparison at that  $N$ , seeds with FR earlier - FR later than ABF (finite beats censored, both censored = tie; \* = exact sign test  $p < 0.05$ ; medians, G and CIs in summary.json / tables.md). Time axis: tau over every save of the run.  
Source results/equal\_budget\_v2/gateway/analysis/summary.json (generated 2026-10-10T08:08:27Z, eqb\_metrics/2, 10000 bootstrap resamples); plot\_synthesis/2.